Newton, quantum mechanics and Dimock–Balaban: a guide through one time dimension

Scope and result, 2026-10-27. A point particle already displays several different questions hidden in the phrase "existence of motion": does a trajectory exist, is it unique, does it continue for all time, and what rule applies at a wall or singularity? This guide connects those questions to a constructive $0+1$ quantum theory and to the multiscale organization of Dimock's exposition of Bałaban's renormalization group. The simple model is not a solution of the four-dimensional Yang–Mills problem.

The detailed quantum-to-Newton proof constructs a family of one-dimensional quantum Hamiltonians with a supplied $\hbar > 0$. For smooth $V \geq 0$ with bounded derivatives of order two and higher, coherent packet means on finite time partitions form a Cauchy family as $\hbar$ and the mesh vanish jointly. The limit is the unique Newton solution; no solution is assumed before the limit. The companion groupoid formulation places the corresponding observable states on the quantum and classical fibers of the tangent groupoid. Here we explain why these constructions work, how they relate to the original sources, and where additional classical examples need new assumptions.

Reading labels below distinguish the actual statements in the cited papers from an analogy or a calculation made here. No priority claim is made for standard ODE, Gaussian, semigroup or semiclassical methods.

1. What must exist, and in what sense?

For a particle in $\mathbb{R}^d$, mass $m > 0$ and conservative potential V, Newton's initial-value problem is

$$\dot{q} = v, \qquad m\dot{v} = -\nabla V(q), \qquad (q(0), v(0)) = (q_0, v_0). \tag{1}$$

It asks for a **trajectory** from initial position and velocity. The endpoints of a path integral ask a different question: prescribe $q(0)$ and $q(T)$ and integrate over intermediate configurations. Even for the harmonic oscillator an endpoint map can become singular at a conjugate time while the initial-value trajectory remains unique.

For $V \in C^1$ with locally Lipschitz ∇V, (1) has a unique local solution, by the usual Cauchy–Lipschitz theorem. Along it,

$$E(q, v) = \tfrac{1}{2}m|v|^2 + V(q), \qquad \frac{d}{dt}E = 0. \tag{2}$$

If V is bounded below, (2) bounds $|v|$. On any finite interval $|q(t)| \leq |q_0| + |t| \sup |v|$, so (q, v) remains in a compact set. The local continuation theorem then gives a unique solution for all positive and negative times. Coercivity of V is a stronger condition than is needed for this finite-time argument.

Smoothness alone does not guarantee global existence. For $m = 1$, $V(q) = -q^4/4$, $q_0 > 0$ and $v_0 = q_0^2/\sqrt{2}$, the zero-energy solution

$$q(t) = \frac{q_0}{1 - q_0 t/\sqrt{2}} \tag{3}$$

escapes to infinity at $t = \sqrt{2}/q_0$. A force singularity can also stop a classical solution: for example, a radial attractive Kepler trajectory can reach $r = 0$ in finite time. A rule for continuing through the singular point is further data.

Regularity also matters for **uniqueness**. For $m = 1$ and $V(q) = -\frac{2}{3}|q|^{3/2}$ near zero, $-V'(q) = \operatorname{sgn}(q)\sqrt{|q|}$. With $(q_0, v_0) = (0, 0)$ there is the stationary solution and, for every $t_0 \geq 0$, the solution that waits at zero until t_0 and then follows $q(t) = (t - t_0)^4/144$, as long as it stays in the displayed neighbourhood. The potential can be cut off smoothly away from

zero and shifted by a constant to be nonnegative and bounded. Its quantum Hamiltonian is then a bounded perturbation of the free self-adjoint Hamiltonian, while its classical initial-value problem is nonunique. Quantum existence by itself therefore does not supply a unique classical branch.

2. A velocity-dependent force: magnetism

In three dimensions the Lorentz force, with a static scalar potential and a static magnetic field, gives

$$\dot{q} = v, \qquad m\dot{v} = -\nabla V(q) + e\, v \times B(q). \tag{4}$$

It is not a force depending on position alone. If ∇V and B are defined and locally Lipschitz on all of $\mathbb{R}^3$, the full phase-space vector field is locally Lipschitz, so local existence and uniqueness still hold. Moreover $v \cdot (v \times B) = 0$, hence the energy (2) is conserved. If V is bounded below, the same speed and position bounds as above prove global existence and uniqueness, even if B is not globally bounded.

For instance, a constant field $B = B_0\hat{z}$ rotates the transverse velocity with cyclotron frequency eB_0/m and leaves the longitudinal velocity constant. A velocity-dependent force is thus not an exception to the ODE theory; its dependence must be included in the phase-space vector field being estimated.

If $B = \nabla \times A$ on $\mathbb{R}^3$, the quantum Hamiltonian is

$$H_{\hbar,A} = \frac{1}{2m}(-i\hbar\nabla - eA(q))^2 + V(q). \tag{5}$$

The kinetic momenta generally do not commute: $[\pi_i, \pi_j] = i\hbar e(\partial_i A_j - \partial_j A_i)$. The positive Brownian-bridge density below does not directly cover this magnetic problem: a magnetic Euclidean representation contains a phase. The present coherent-packet theorem is proved for the scalar Hamiltonian in (11), not for (5).

3. A wall is an impact rule, not a smooth force

Take the half-line $q \geq 0$ and a free particle starting at $q_0 > 0$ with $p_0 < 0$. For an ideal **elastic** wall, the prescribed impact law is $p(t_*+) = -p(t_*-)$ at $t_* = -mq_0/p_0$. The motion is

$$q(t) = |q_0 + p_0 t/m|, \qquad p(t) = p_0 \operatorname{sgn}(q_0 + p_0 t/m) \quad (t \neq t_*). \tag{6}$$

Position is continuous; momentum has a jump. Equation (1) with a finite smooth force does not describe that jump. Nonpenetration alone does not select the elastic law: a restitution coefficient or another contact rule would be additional data.

For a smooth wall in several dimensions, the elastic specular rule reverses the normal component and retains the tangential component: $v_+ = v_- - 2(v_- \cdot n)n$. This requires a specified unit normal and a well-defined collision. Corners, grazing contacts and an accumulation of impacts require separate analysis; one should not infer their uniqueness merely from the local smooth-force theorem.

A mathematically defined quantum impenetrable wall can be represented by the Dirichlet Laplacian

$$H_{\hbar,D} = -\frac{\hbar^2}{2m}\frac{d^2}{dq^2}, \qquad D(H_{\hbar,D}) = H^2(0,\infty) \cap H_0^1(0,\infty). \tag{7}$$

It is self-adjoint and gives a unitary group. Its Euclidean kernel is

$$K_D(T; x, y) = k_{\hbar,T}(x - y) - k_{\hbar,T}(x + y), \quad k_{\hbar,T}(r) = \sqrt{\frac{m}{2\pi\hbar T}}e^{-mr^2/(2\hbar T)}, \quad x, y > 0. \tag{8}$$

This is the method of images: the Euclidean reference process is Brownian motion killed at the boundary. The quantum real-time wall reflects, but killed Euclidean paths should not

themselves be read as real-time bouncing trajectories. Section 8 proves a quantitative coherent-packet position limit through the bounce.

A subtle domain issue is important: the symmetric differential operator $-i\hbar\, d/dq$ on the half-line with Dirichlet boundary has unequal deficiency indices and no self-adjoint extension on $L^2(0,\infty)$. The Hamiltonian (7) is nevertheless self-adjoint. A current or a quadratic-form expectation involving $-i\hbar d/dq$ can be studied, but it should not be presented as the expectation of a self-adjoint half-line momentum operator. See Al-Hashimi and Wiese, arXiv:2103.01715 (abstract) for the boundary momentum problem.

4. A finite Euclidean lattice and one exact block

Return to the whole line and choose endpoints x, y, duration $T = u + v$, and action unit $\hbar > 0$. The free Euclidean kernel is $k_{\hbar,T}$ from (8). Its semigroup identity, refined at time u, is

$$k_{\hbar,T}(y-x) = \int_{\mathbb{R}} k_{\hbar,u}(z-x)k_{\hbar,v}(y-z)\,dz. \tag{9}$$

Completing the square gives the **conditional** law of the intermediate coordinate:

$$k_{\hbar,u}(z-x)k_{\hbar,v}(y-z) = k_{\hbar,T}(y-x)\sqrt{\frac{a}{2\pi\hbar}}\,e^{-a(z-\mu)^2/(2\hbar)}, \quad a = \frac{mT}{uv}, \quad \mu = \frac{vx+uy}{T}. \tag{10}$$

The normalization in (10) matters: an action penalty is not yet a probability bound until the full conditional integral is divided out. The inserted point has mean μ and variance $\hbar/a = \hbar uv/(mT)$.

For a nonnegative potential, the finite-time weight adds $\exp[-\sum_j \delta_j(V(q_j)+V(q_{j+1}))/(2\hbar)]$. As the mesh tends to zero, the Brownian-bridge expectation converges by dominated convergence. With $Z_\hbar(x,y)$ denoting that expectation, the kernel of the Euclidean semigroup is

$$K_\hbar(T;x,y) = k_{\hbar,T}(y-x)Z_\hbar(x,y), \qquad H_\hbar = P^2/(2m) + V(Q). \tag{11}$$

Feynman–Kac and the spectral theorem then provide the real-time $U_\hbar(t) = e^{-itH_\hbar/\hbar}$. The Brownian measure and these standard operator theorems are inputs; the construction is not a lattice-only derivation of Wiener measure.

There is also a sign distinction. The Euclidean stationary equation for $\int(\frac{1}{2}m\dot{q}^2 + V)\,dt$ is $m\ddot{q} = +V'(q)$. The real-time equation for $\int(\frac{1}{2}m\dot{q}^2 - V)\,dt$ is $m\ddot{q} = -V'(q)$. The positive-weight minimizer cannot be relabelled as a Newton trajectory without another argument.

5. Reading Dimock and Bałaban from this example

Dimock's two papers develop Bałaban's method for a **three-dimensional lattice scalar ϕ^4 theory**, with ultraviolet stability of its partition function as the illustrative problem. Their scalar results do not by themselves establish the Yang–Mills mass gap. The following references are to specific statements and passages, not a claim that their full proofs have been audited.

1. **Keep the complete density.** In Dimock I, §2.1, equations (15), (26)–(28), Q averages a fine field over a block; the next density is obtained by integrating the previous density against a normalized Gaussian kernel, then rescaling. Equation (28) preserves the total integral. The $0+1$ operation (10) instead eliminates a time coordinate conditional on retained coordinates. Our density $\rho_n = \mathbb{E}_0[\rho \mid \mathcal{F}_n]$ is relative to a Brownian-bridge reference and satisfies $\rho_n = \mathbb{E}_0[\rho_{n+1} \mid \mathcal{F}_n]$. This is an exact tower, but it is not literally Dimock's noisy block-average map, nor does it contain his rescaling.

2. **A minimizer is only the center of an integral.** Completing the free square in (10) produces a background point μ *and* its covariance $\hbar/a$ and normalization. Dimock I, §§2.2–2.3, likewise separates a background and Gaussian fluctuation in its free flow. With interactions, the analogue is a full conditional law around a generally nonlinear minimizer, not merely the minimizer itself.

3. **Generated interactions must travel to the next scale.** For one inserted vertex, with $w = T/2$, define

$$A_\hbar(\mu) = -\hbar \log \left[\sqrt{\frac{a}{2\pi\hbar}} \int e^{-[a(z-\mu)^2/2 + wV(z)]/\hbar} dz \right]. \tag{12}$$

This entire function enters the retained kernel. If $V(z) = \lambda z^4/4$, differentiating at $\lambda = 0+$ gives

$$\partial_\lambda A_\hbar(\mu)|_{0+} = \frac{w}{4} \left[\mu^4 + 6\mu^2 \frac{\hbar}{a} + 3 \left(\frac{\hbar}{a} \right)^2 \right]. \tag{13}$$

Even one integration generates a quadratic term and a constant in addition to the quartic term. The derivative is not a claim of convergence of the whole perturbation series. Dimock I, §4.1, equations (176)–(180), and Theorem 14 propagate a much richer representation: a normalization, mass and energy terms, a quartic term and a controlled polymer remainder, under the theorem's small-coupling and sufficiently-large-block hypotheses. The nonlinear-cell calculation gives nonperturbative bounds for (12).

4. **Small fields are not the whole integral.** Dimock I, equations (172)–(175), introduces characteristic functions for the small-field analysis. Dimock II, introduction and §3.3, restores the full expansion by summing over histories of regions; its Theorem 3.1 includes large-field and boundary terms and normalization, under its stated block-size, mass and running-coupling conditions and with the energy and mass counterterms selected in Part I. The elementary identity $\prod_b [\chi_b + (1 - \chi_b)] = 1$ illustrates why every pattern of good and bad blocks is retained. Dimock's good-region tests involve fields, derivatives and nested regions; they are not identical to a bound on one midpoint fluctuation. In the one-cell law, strong convexity yields the normalized bound

$$\mathbb{P}(|Z - z_*| \geq r \mid \text{retained endpoints}) \leq 2e^{-\kappa r^2/(2\hbar)}, \tag{14}$$

provided the conditional potential has second derivative at least $\kappa > 0$. The integration over its complement is controlled, not deleted.

5. **Locality of correlated fluctuations must be earned.** Dimock I, §2.4 and Lemma 6, constructs a Green function using local inverses and a convergent random-walk expansion; §4.5 introduces weakening parameters before the cluster expansion. Dimock II, §3.13 and Lemmas 3.15–3.17, localizes the interaction, remainder and existing boundary terms; §3.14 and Lemma 3.18 then perform a cluster expansion with holes. Locality in a correlated fluctuation field is not automatically locality in independent Gaussian variables. In the nearest-neighbour time chain, nonadjacent eliminated vertices are conditionally independent **for one halving**. That elementary fact does not prove a multi-scale polymer estimate in three or four dimensions.

The continuity of the normalized 0+1 density tower follows here from conditional expectation and an already constructed Brownian bridge. The difficult field-theory task is to retain useful **uniform, local bounds** on effective interactions, large-field regions and boundaries as the cutoff is removed. An exact composition identity alone is not that estimate.

6. From the quantum blocks to a Newton path

Fix a Gaussian coherent packet centered at (q_0, p_0) with position and momentum standard deviations of order $\sqrt{\hbar}$. At each time slice first kick by the potential and then drift freely:

$$K_{\hbar,s} = e^{-isV(Q)/\hbar}, \qquad D_{\hbar,s} = e^{-isP^2/(2m\hbar)}, \qquad T_{\hbar,s} = D_{\hbar,s} K_{\hbar,s}. \tag{15}$$

For finite products the exact conjugations are

$$K_{\hbar,s}^{*} P K_{\hbar,s} = P - sV'(Q), \qquad D_{\hbar,s}^{*} Q D_{\hbar,s} = Q + sP/m. \tag{16}$$

If $|V''| \le L$, the standard deviation of $V'(Q)$ is at most L sd(Q). A discrete Gronwall estimate consequently bounds both position and momentum spreads by $C_T\sqrt{\hbar}$, uniformly over partitions of $[0, T]$.

Let $z_{\hbar,\pi}(t)$ interpolate the means after the finite cells. The spread bound and (16) imply the force-balance residual

$$\|\dot{z}_{\hbar,\pi} - \mathcal{F}(z_{\hbar,\pi})\| \le C_T(\sqrt{\hbar} + |\pi|), \qquad \mathcal{F}(q,p) = (p/m, -V'(q)). \tag{17}$$

Comparing two independently chosen $\hbar$'s and meshes, and applying Gronwall, shows that these interpolants form a Cauchy family. Its limit is the unique solution of (1), with

$$\sup_{t \le T} \|z_{\hbar,\pi}(t) - z(t)\| \le C_T(\sqrt{\hbar} + |\pi|). \tag{18}$$

The detailed proof gives the weighted phase-space norm, constants, arbitrary-partition product limit and domains. The statement is about means; it does not assert that finite-product propagators converge uniformly in operator norm along arbitrary joint schedules.

Cauchy–Lipschitz constructs the integral equation directly, usually by a contraction on a short interval. The route above constructs quantum operators first and their classical limit afterward. Both use the same Lipschitz stability of the force and Gronwall uniqueness. The extra quantum structure relates the solution to a family of states; it does not weaken the regularity needed here or derive $\hbar > 0$ from Newton's law. Hepp's 1974 coherent-state theorem is established prior work on quantum correlation functions and classical orbits (abstract and introduction read).

7. The tangent groupoid and the separate scales

For $M = \mathbb{R}$, the tangent groupoid has positive fibers $M \times M$ and a zero fiber TM. Its observable C^*-algebra is a continuous field, with

$$\mathcal{A}_{\epsilon>0} \simeq \mathcal{K}(L^2(\mathbb{R})), \qquad \mathcal{A}_0 \simeq C_0(T^*\mathbb{R}). \tag{19}$$

After choosing an action unit S_* and identifying $\hbar = \epsilon S_*$, the state estimate in the groupoid companion connects the coherent quantum states to evaluation on the constructed classical trajectory. This uses the established continuous-field and strict quantization framework of Cariñena et al. and Landsman–Ramazan (abstracts read).

Three operations remain distinguishable: ϵ labels a geometric deformation; the time mesh controls a finite product; and blocking integrates over intermediate variables. Choosing a joint schedule ties their parameters, but does not make geometric zoom identical to Wilsonian integration or prove that the physical action unit runs.

The subsequent dispersion construction extends the packet argument to the positive-energy relativistic Hamiltonian and controls its nonrelativistic limit. Its free spectral Dirichlet wall is a separate nonlocal boundary model.

8. A quantum packet through the elastic bounce

The wall example can be related to quantum theory without applying the smooth-force theorem at $q = 0$. Let $q_0 > 0$, $p_0 < 0$, $\omega > 0$, and let $g_\hbar$ be the normalized full-line Gaussian of position mean q_0, momentum mean p_0, and variances $\hbar/(2m\omega)$ and $\hbar m\omega/2$. If R is reflection, $(Rg)(x) = g(-x)$, define the normalized odd state

$$\phi_\hbar = \frac{g_\hbar - Rg_\hbar}{\sqrt{2(1-\gamma_\hbar)}}, \qquad \gamma_\hbar = \langle g_\hbar, Rg_\hbar \rangle = \exp\left[-\frac{m\omega q_0^2 + p_0^2/(m\omega)}{\hbar}\right]. \tag{20}$$

The corresponding Dirichlet half-line state is $\psi_\hbar(q) = \sqrt{2}\,\phi_\hbar(q)$ for $q > 0$. Odd extension intertwines its evolution with free evolution on the full line.

Write $a(t) = q_0 + p_0 t/m$, and let $g_{\hbar,t}$ be the freely evolved Gaussian. Its position variance is

$$\sigma_q(t)^2 = \frac{\hbar}{2m\omega}(1 + \omega^2 t^2). \tag{21}$$

Because $||x| - |a|| \le |x - a|$ and reflection commutes with the free evolution,

$$\left|\langle Q\rangle_{\psi_\hbar(t)} - |a(t)|\right| \le \sqrt{\frac{2}{1-\gamma_\hbar}}\,\sigma_q(t). \tag{22}$$

Indeed the half-line position expectation equals the full-line expectation of $|X|$ in the odd state; apply the triangle inequality to the two Gaussian terms. Thus position converges uniformly on every bounded time interval, **including the impact time**, to the elastic classical trajectory (6).

Away from the impact, the corresponding probability-current momentum also converges to the incoming or reflected value. More precisely, if $|a(t)| \ge \eta > 0$, with $\sigma_p = \sqrt{\hbar m\omega/2}$,

$$\left|\int_0^\infty \overline{\psi_\hbar}(-i\hbar\partial_q)\psi_\hbar\,dq - p_0 \operatorname{sgn} a(t)\right| \le \sqrt{\frac{2}{1-\gamma_\hbar}}\left(\sigma_p + \frac{2|p_0|\sigma_q(t)}{\eta}\right). \tag{23}$$

For the proof, the half-line integral is the full-line expectation of $\operatorname{sgn}(X)P$ in the odd state. On the Gaussian, $\|(\operatorname{sgn} X\,P - \operatorname{sgn} a\,p_0)g_t\|$ is at most $\sigma_p + 2|p_0|\mathbb{P}(aX < 0)^{1/2}$; Chebyshev bounds the last probability by $\sigma_q(t)^2/a(t)^2$. The reflected Gaussian obeys the same estimate. For each fixed $T, \eta > 0$ independent of $\hbar$, (23) proves uniform convergence over $|t| \le T$ with $|a(t)| \ge \eta$. It gives no uniform bound in an $\hbar$-dependent impact layer $|a(t)| \sim \sqrt{\hbar}$ or momentum value at t_*. The expression in (23) is a real current form on this domain, not an expectation of a self-adjoint half-line momentum.

The bound deliberately excludes t_*. A family of continuous quantum current expectations cannot converge uniformly across t_* to the discontinuous classical momentum in (6). Position convergence and momentum convergence therefore have different valid topologies.

9. Consequence for STATE

The requested didactic bridge now includes explicit source locations, the normalized one-cell operation, generated terms and the limitation of the analogy. It distinguishes classical smooth-force existence, continuation, nonuniqueness, magnetic forces and an imposed hard-wall impact. The wall packet calculation gives a separate rigorous example of a classical limit through an impact. The supplied action parameter, the independent action-necessity problem and the higher-dimensional field-theory estimates remain open in their respective scopes. The later canonical central-limit calculation identifies a Gaussian preparation sector under additional copying and cusp-compatibility assumptions; its floor is not identified with the quantum phase constant.

Targeted advisor check (GPT-6.1 Sol, 2026-10-27). The impact packet estimates and their convergence domains, the magnetic continuation argument, and the cited Dimock source locations were checked against the written calculation and selected source passages. This is an internal model check, not independent peer review of the imported ODE, operator, probability or field-theory theorems.